

Observed program direction**Complementary perspective****Interpretation boundary****Thymus**

DNA repair, cytoskeletal control,
and lymphoid-stromal
interaction lower

Adaptive immune decline includes
weaker cell coordination and
niche-associated programs.

Bulk tissue cannot separate
thymocytes from stromal-cell
abundance or state.

Skin

Chromatin, DNA repair, Hedgehog,
sphingolipid, and junction
programs lower

Barrier stress is accompanied by
a broad maintenance and
communication decrease.

Gravity, site, recovery, strain,
and duration modify the
pooled response.

Liver

MHC class II presentation and
T-cell receptor signaling
lower

Metabolic disruption is
accompanied by lower adaptive
immune communication.

Immune-cell abundance and
signaling state are not separable
in bulk RNA-seq.

Spleen

T-cell receptor, neutrophil
degranulation, and C-type
lectin programs lower

Adaptive suppression extends to
lower innate effector and
pathogen-sensing programs.

Transcriptomic programs are not
direct functional assays of
immune activity.

Vector synthesis of retained pathway directions; columns organize interpretation and do not assert causality.